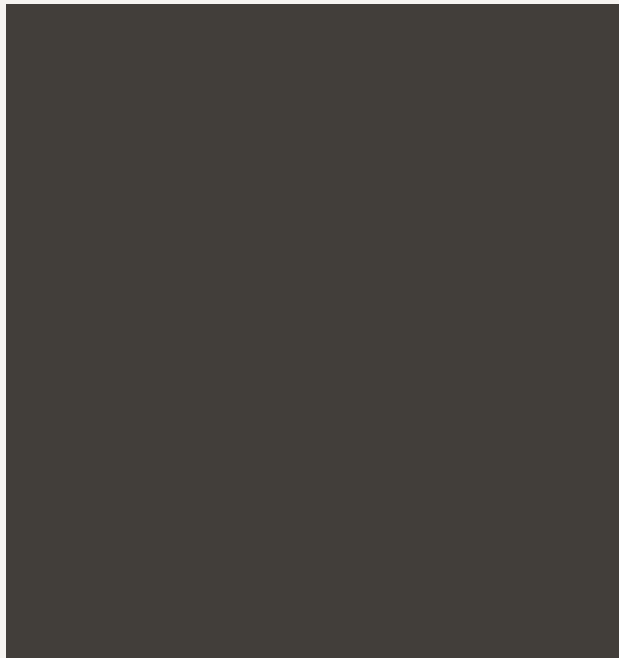


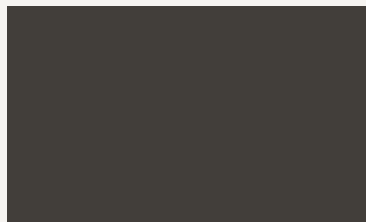


Appointment Management System





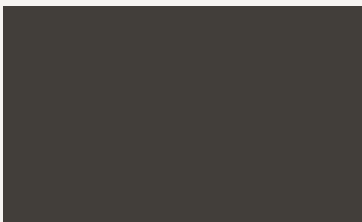
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Introduction & Project Description

The Appointment Management System is used to manage patients, doctors, employees, and appointments in a healthcare facility. It helps in booking and organizing appointments in an easy and clear way by storing all information in a structured database. The system reduces appointment conflicts, improves staff coordination, and supports smoother and more organized hospital operations.



Scenario



An appointment management system is used to organize patients, doctors, appointments, departments, and employees within a healthcare facility .

Each patient is described by patient ID, full name, email, phone number, and birth date. The patient ID is unique for each patient .

Each doctor is described by doctor ID, name, specialty, years of experience, and phone number. The doctor ID is unique for each doctor .

Each department is described by department ID, department name, and number of doctors in the department. The department ID is unique for each department .

Each employee is described by employee ID, name, job title, email, and working hours. The employee ID is unique for each employee .



Scenario



Each appointment is described by appointment ID, appointment date, start time, end time, and status, and references a patient and a doctor. The appointment ID is unique for each appointment.

Each doctor can handle multiple appointments, while each appointment must be assigned to exactly one doctor .

Each patient can book multiple appointments, while each appointment must belong to exactly one patient .

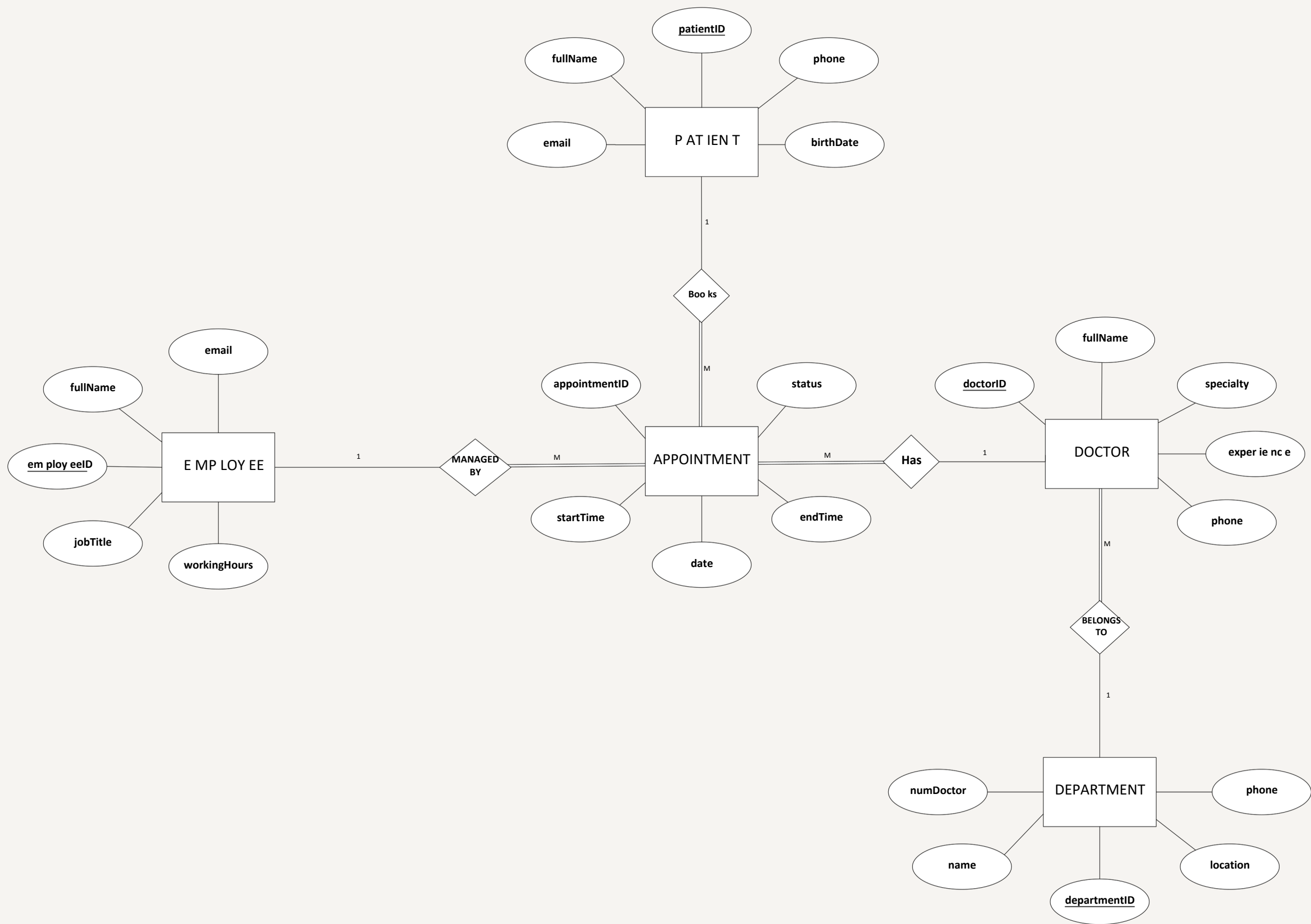
Each department includes multiple doctors, while each doctor belongs to exactly one department .

Each employee can manage or update multiple appointments, while each appointment must be supervised by exactly one employee .

When a new appointment is created, the system checks for time conflicts with the doctor 's existing appointments, and if a conflict is found, the appointment is rejected or rescheduled .



ER Diagram



Schema Diagram

Department

numDoctor	name	<u>departmentID</u>	location	phone
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Doctor

<u>doctorID</u>	fullName	specialty	experience	phone	departmentID
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Employee

workingHours	jobTitle	<u>employeeID</u>	fullName	email
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Patient

email	fullName	<u>patientID</u>	phone	birthDate
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Appointment

appointmentID	startTime	date	endTime	status	doctorID	employeeID	patientID
---------------	-----------	------	---------	--------	----------	------------	-----------

SQL Code

```
queries.sql + 44nvvegyn
1 ALTER TABLE DEPARTMENT ADD notes VARCHAR(255);
2
3 CREATE TABLE DOCTOR (
4     doctorID INT PRIMARY KEY,
5     name VARCHAR(100) NOT NULL,
6     specialty VARCHAR(100),
7     experience INT,
8     phone VARCHAR(20),
9     deptID INT,
10    FOREIGN KEY (deptID) REFERENCES DEPARTMENT(departmentID)
11 );
12
13 CREATE TABLE PATIENT (
14     patientID INT PRIMARY KEY,
15     fullName VARCHAR(100) NOT NULL,
16     email VARCHAR(100),
17     phone VARCHAR(20),
18     birthDate DATE
19 );
20
21 ALTER TABLE PATIENT ADD age INT;
22
23 CREATE TABLE EMPLOYEE (
24     employeeID INT PRIMARY KEY,
25     name VARCHAR(100) NOT NULL,
26     jobTitle VARCHAR(50),
27     email VARCHAR(100),
28     workingHours VARCHAR(50)
29 );
30
31 CREATE TABLE APPOINTMENT (
32     appointmentID INT PRIMARY KEY,
33     date DATE NOT NULL,
34     startTime TIME NOT NULL,
35     endTime TIME NOT NULL,
36     status VARCHAR(20),
37     patientID INT,
38     doctorID INT,
39     employeeID INT,
40     FOREIGN KEY (patientID) REFERENCES PATIENT(patientID),
41     FOREIGN KEY (doctorID) REFERENCES DOCTOR(doctorID),
42     FOREIGN KEY (employeeID) REFERENCES EMPLOYEE(employeeID)
43 );
44
45 INSERT INTO DEPARTMENT (departmentID, name, numDoctor, location, phone) VALUES
46 (101, 'Cardiology', 5, 'Building A, Floor 1', '012345601'),
47 (102, 'Neurology', 3, 'Building A, Floor 2', '012345602'),
48 (103, 'Pediatrics', 4, 'Building B, Floor 1', '012345603'),
49 (104, 'Orthopedics', 4, 'Building B, Floor 2', '012345604'),
50 (105, 'Dermatology', 4, 'Building C, Floor 1', '012345605'),
51 (106, 'Ophthalmology', 2, 'Building C, Floor 2', '012345606'),
52 (107, 'Emergency Room', 10, 'Building D, Floor 1', '012345607');
53
54
55 INSERT INTO DOCTOR VALUES (201, 'Haitham Ahmed', 'Cardiology', 15, '055000111', 101);
56 INSERT INTO DOCTOR VALUES (202, 'Tamim Mohammed', 'Neurology', 8, '055000222', 102);
57 INSERT INTO DOCTOR VALUES (203, 'Noah Yousef', 'Pediatrics', 12, '055000333', 103);
58 INSERT INTO DOCTOR VALUES (204, 'Sadeem Saud', 'Orthopedics', 20, '055000444', 104);
59 INSERT INTO DOCTOR VALUES (205, 'Aleen Meshal', 'Dermatology', 5, '055000555', 105);
60 INSERT INTO DOCTOR VALUES (206, 'Yara Omar', 'Ophthalmology', 10, '055000666', 106);
61
62
63 INSERT INTO PATIENT (patientID, fullName, email, phone, birthDate) VALUES
64 (303, 'Fatima Ali', 'fatima@mail.com', '056000223', '1992-03-15'),
65 (304, 'Amal Mansour', 'amal@mail.com', '056000224', '1998-07-22'),
66 (305, 'Ahlam Fahad', 'ahlam@mail.com', '056000225', '1980-12-05'),
67 (306, 'Ahmed Hassan', 'ahmed@mail.com', '056000226', '1995-01-30'),
68 (307, 'Mohammed Salem', 'mohammed@mail.com', '056000227', '1988-11-12');
69
70
71 INSERT INTO EMPLOYEE VALUES (403, 'Azzam Ali', 'Coordinator', 'azzam@clinic.com', '08:00 - 16:00');
72 INSERT INTO EMPLOYEE VALUES (404, 'Yasmin Ali', 'Secretary', 'yasmin@clinic.com', '09:00 - 17:00');
73
74
75 INSERT INTO APPOINTMENT VALUES (1, '2026-06-01', '09:00:00', '09:30:00', 'Completed', 303, 201, 403);
76 INSERT INTO APPOINTMENT VALUES (2, '2026-06-01', '10:00:00', '10:30:00', 'Scheduled', 304, 202, 403);
77 INSERT INTO APPOINTMENT VALUES (3, '2026-06-02', '11:00:00', '11:30:00', 'Pending', 305, 203, 404);
78 INSERT INTO APPOINTMENT VALUES (4, '2026-06-03', '13:00:00', '13:30:00', 'Scheduled', 306, 204, 403);
79 INSERT INTO APPOINTMENT VALUES (5, '2026-06-04', '15:00:00', '15:30:00', 'Confirmed', 307, 205, 404);
80
81
82 UPDATE APPOINTMENT SET date = '2026-07-20', status = 'Rescheduled' WHERE appointmentID = 1;
83
84
85 DELETE FROM APPOINTMENT WHERE appointmentID = 3;
86
87
88 SELECT A.appointmentID, P.fullName AS Patient_Name, D.name AS Doctor_Name, A.date, A.status
89 FROM APPOINTMENT A
90 JOIN PATIENT P ON A.patientID = P.patientID
91 JOIN DOCTOR D ON A.doctorID = D.doctorID
92 ORDER BY A.date ASC;
93
94
95 SELECT DEP.name AS Department, COUNT(DOC.doctorID) AS Total_Doctors
96 FROM DEPARTMENT DEP
97 LEFT JOIN DOCTOR DOC ON DEP.departmentID = DOC.deptID
98 GROUP BY DEP.name
99 ORDER BY Total_Doctors DESC;
100
```

Output

Output:

183 ms

appointmentID	PatientName	DoctorName	date	status
1	Fatima Ali	James Wilson	2026-06-01	Completed
2	Amal Mansour	Mary Johnson	2026-06-01	Scheduled
3	Ahlam Fahad	Robert Brown	2026-06-02	Confirmed
4	Ahmed Hassan	Linda Davis	2026-06-03	Scheduled
5	Mohammed Salem	Michael Miller	2026-06-04	Pending
6	Saad Khalid	Susan Wilson	2026-06-05	Scheduled

name	DoctorCount
Cardiology	1
Neurology	1
Pediatrics	1
Orthopedics	1
Dermatology	1
Ophthalmology	1



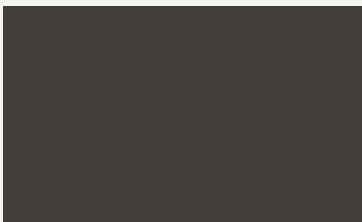
Project Challenges

Identifying entity relationships

Defining relationship cardinality

Converting ER Diagram to Schema

Maintaining data consistency





Conclusion

The system improves scheduling efficiency and prevents appointment conflicts

THANK YOU

